RENÉ KUHM

Full Stack Developer & AI Engineer — AI, Automation & Observability

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8+ years
Software engineering & automation (2018–present)

~1,600 CI tests
Verifiable quality & architecture

9+ years
Mission-critical FTTH/GPON & NOC operations

6 systems with public code
E-commerce, SaaS RAG, CLI/MCP, Rust, password manager

PROFESSIONAL SUMMARY

Full-stack developer **building systems end to end since 2018**: architecture, code, automated testing, CI/CD and observability, from multi-tenant APIs to autonomous agent orchestrators. Behind that sits **9+ years operating real FTTH/GPON networks** in the field and at the headend: I build software for infrastructure I know from the inside.

TECHNICAL SKILLS

- **Full-stack, backend & systems development:** TypeScript (strict), Python, Node.js, NestJS, FastAPI, Next.js, React, Flutter, Rust, multi-tenant architecture, systems design.
- **Applied AI, autonomous agents & automation:** Model Context Protocol (MCP), LangGraph, CrewAI, AutoGen, BeeAI, RAG with pgvector, spec-driven orchestration with persistent context, n8n, LLM integration.
- **Data, DevOps, Linux & observability:** PostgreSQL (SQL / pgvector), Docker, Docker Compose, Linux (NixOS, systemd, advanced administration), GitHub Actions (CI/CD), Prometheus, Grafana, OpenTelemetry, Sentry, Playwright (E2E), Cloudflare, Vercel.
- **Cybersecurity & applied cryptography:** Argon2id key derivation, AES-256-GCM authenticated encryption, revocable HttpOnly JWT sessions, rate limiting, audit trails and threat detection via Syslog.
- **Telecommunications, NOC/SOC & ISP infrastructure:** FTTH/GPON, ADSL, OLT/ONT headend (SmartOLT / MikroWISP), optical RX drift and flapping detection, alert correlation, TCP/IP and outside plant (OSP).

PROFESSIONAL EXPERIENCE

Founder · Full-Stack Developer & Automation Architect — TecnoDespegue Jan 2018 – Present (8+ yrs) · Remote

- **Full-stack and backend development:** design and put into production web applications, SaaS platforms, REST APIs and internal systems with TypeScript, Next.js, React, Node.js, NestJS, Python, FastAPI, PostgreSQL and Docker, owning the full delivery lifecycle.
- **NOC/SOC platforms and observability for ISPs:** monitoring and NMS-integrated solutions running metric pollers, proactive detection of optical RX drift, temperature and flapping, Syslog ingestion, alert correlation with cooldown and automated notifications via Webhooks/Telegram on top of Prometheus, Grafana, OpenTelemetry and Sentry.
- **Application security and cryptography:** encrypted credential storage, revocable HttpOnly JWT sessions, audit trails, rate limiting and offline-first encryption with Argon2id and AES-256-GCM across multiple products.
- **Code quality:** standardize delivery with test suites of up to ~1,600 automated tests in CI and maintain six of my own systems with public code on GitHub.

Telecommunications Technical Specialist · FTTH/GPON & Headend Operations — COSPEC Ltda. Jan 2017 – Apr 2026 (9 yrs 4 mo) · La Pampa

- **Headend and technology migration at scale:** assembled and commissioned the FTTH headend and telecom racks, and served as a key technician in the complete network transition from copper/ADSL to FTTH/GPON — trunk deployment, urban optical ring, rural extension and wireless links.
- **End-to-end diagnostics and proven mass execution:** incidents resolved from the subscriber's premises through outside plant (splicing, splitters, closures) up to the headend under SLA; ~2,500 FTTH installations plus an equivalent ADSL volume, with signal-quality metrics that inspired FTTH-Copilot's architecture.

PROJECTS

FTTH-Copilot — flagship project `github.com/Rene-Kuhm/FTTH-Copilot`
Python · FastAPI · TypeScript · Syslog · NMS · JWT · 2026–present · concept & field research since 2024
Multi-tenant NOC/SOC platform for ISPs covering **17 documented OLT vendors** and 22 simulated device families, with ~1,600 automated tests in CI. Integrates SmartOLT/MikroWISP, RX-drift pollers and alert deduplication.

AcuaMarina Cerámica `github.com/Rene-Kuhm/acuamarina-ceramica`
E-commerce · TypeScript · Backend + Admin · MercadoPago · Railway · public since Oct 2025
Full-stack e-commerce for a real ceramics shop: backend, admin dashboard, frontend, MercadoPago payments, deployed on Railway.

flow-engineering `github.com/Rene-Kuhm/flow-engineering`
Python · CLI · MCP Server · public since Jun 2026
Spec-driven orchestrator for AI-assisted software development: persistent context, automated verification flows and a read-only Model Context Protocol (MCP) server.

Vaulta `github.com/Rene-Kuhm/vaulta`
Flutter · Argon2id · AES-256-GCM · Offline-first · public since Mar 2026
Cross-platform (Android/Windows) password manager with an offline-first, zero-knowledge architecture: key derivation resistant to hardware attacks and authenticated encryption.

campusmind-saas `github.com/Rene-Kuhm/campusmind-saas`
Next.js · NestJS · PostgreSQL (pgvector) · Docker · Playwright · public since Dec 2025
Full-stack AI study SaaS platform: vector search and RAG over the user's own material, deployment with Docker Compose and end-to-end testing with Playwright.

redragon-streamdeck-linux `github.com/Rene-Kuhm/redragon-streamdeck-linux`
Rust · TypeScript · Linux systemd · USB/HID · public Jan–Sep 2026
Low-level open-source driver and control panel for the Redragon SS-550 Stream Deck on Linux: systems programming in Rust, systemd daemons and direct USB/HID communication.

EDUCATION & CERTIFICATIONS

- **IBM Full Stack Software Developer Professional Certificate** — IBM / Coursera (Aug 2026 — 15/15 courses + capstone): cloud-native, microservices, Docker, CI/CD.
- **Model Context Protocol (MCP): Advanced Topics & AI Assisted Development** — Anthropic (2026): agentic architectures and context standards for LLMs.
- **Agentic AI with LangGraph, CrewAI, AutoGen and BeeAI** — IBM / Coursera (Aug 2026), part of the IBM RAG and Agentic AI Professional Certificate.
- **Networking Fundamentals & Network Access** — Packt (2026); **Technical Support Fundamentals** — Google (2026).
- **Python Essentials 1** — Cisco Networking Academy (2025); **Data Analysis with Python** — freeCodeCamp (2024).
- **JavaScript and React JS Software Development** — Coderhouse (2022–2023).
- **Languages:** Spanish (native) · English — B2 in technical reading and writing; conversational fluency in progress.